Beyond Spatial Accuracy: Diagnosing Persistent Orientation Failures in Vision–Language Models

Anonymous WACV Datasets Track submission

Paper ID *****

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Supplementary Material

1. Exact Prompts, Decoding, and Parsing

VSR plain prompt (identical for 2B and 7B, all conditions; ‘{statement}’ is the VSR caption):

```
Look at the image carefully.

Statement: ”{statement}”

Is this statement true or false?

Answer with exactly one word:
True or False.
```

2B structured-decomposition prompt (2B only; single run):

```
Look carefully at the image.

Statement:
”{statement}”

Determine whether the statement is true
by doing the following:
1. Identify the subject object.
2. Identify the reference object.
3. Determine the position and
   orientation of the subject relative
   to the reference object.
4. Pay careful attention to left/right,
   front/back, viewpoint, and
   orientation.
5. Decide whether the stated spatial
   relationship matches the image.

Answer with exactly one word:
True or False.
```

SITE multiple-choice prompt (replicated from the official lmms-eval task sitebench/utils.py):

```
# image examples (one <image> token per visual):
<image>...
Question: {question}
Options:
A: {opt0}
B: {opt1}
...
Give me the answer letter directly.
The best answer is:

# video examples (pre-prompt):
Select the best answer to the following
multiple-choice question based on the
video. Respond with only the letter
of the correct option.
Question: {question}
Options:
A: ...
B: ...
...
Give me the answer letter directly.
The best answer is:
```

Decoding. All generations are greedy: do_sample=False, temperature 0, top-p 1.0. max_new_tokens: 5 for the 2B wrapper, 128 for 7B VSR runs, 16 for SITE (change validated parse-identical on 125/125 held-out outputs before adoption; recorded in the SITE run metadata).

Parsing. VSR outputs are parsed with an anchored case-insensitive regex (src/evaluation/parser.py): strips prefixes (‘the answer is’, ‘answer:’, ‘response:’, ‘output:’) and trailing punctuation; accepts exact ‘true/yes/correct’ or ‘false/no/incorrect’ forms; substring fallback only when exactly one of ‘true’/’false’ appears. Invalid outputs are counted as wrong, never guessed; invalid rate is 0 on all reported runs. SITE outputs use the official parse_multi_choice_response (letter); unparseable responses are counted incorrect and reported.

2. Training and Adapter Configurations

All LoRA conditions share: r=8, α=16, dropout 0.05, AdamW lr 10^{-4}, weight decay 0.01, linear warmup 10% of steps, 2 epochs, bf16, gradient checkpointing, gradient clipping 1.0, vision tower frozen. Adapter configs are committed at checkpoints/*/final/adapter_config.json.

Manifest construction. General: 2,000 statements sampled from the VSR train split. Targeted: orientation-over-sampled subset of the same split. Hard-negative: the orientation block of the general manifest is replaced by audited-clean originals plus paired hard negatives (facing↔facing away from; parallel↔perpendicular) with flipped labels; every included example was audited for label correctness (451

Condition	LoRA targets	Trainable	Batch	Manifest
2B General	LM q/k/v/o	~2.3M	microbatch 4×4	general (2,000)
2B Targeted	LM q/k/v/o	~2.3M	same	targeted (2,000)
7B LM-only	LM q/k/v/o	~0.3M	1	general (2,000)
7B HardNeg	LM q/k/v/o	~0.3M	1	hard-negative (2,000)
7B Projector	visual.merger.mlp.0/2	152K	1	general (2,000)
7B Vision+Proj	merger + blocks 24–31	1.46M	1	general (2,000)

Table 1. LoRA conditions and their target modules.

train orientation examples audited: 402 clean, 28 excluded, 21 original-only). 2B runs split 5% of the manifest for validation (seed 42). All manifests derive exclusively from the VSR train split; all evaluations use test only.

3. Probe Methodology and Full Tables

Features. Qwen2-VL-7B-Instruct, vision tower frozen. ViT patch embeddings (1280-d) and post-merger features (3584-d), mean-pooled per image (ungrounded) or per grounding box (grounded). Grounding boxes from frozen Qwen2-VL grounding (94.6% of 647 orientation images yield both boxes; failures are mostly genuinely absent objects), rescaled to pixels; region features mean-pooled over patch centers inside each box.

Tasks. T1 facing vs. facing away (test n=103, majority 64.2%); T2 parallel vs. perpendicular (n=34, majority 64.7%); T3 4-way (n=137, majority 49.6%). Training on audited-clean VSR train; 5-fold CV; no test examples in training.

Readouts. Linear logistic regression; MLP (256 hidden, ReLU, Adam, early stopping on CV); patch-vote (per-patch linear probe, majority vote over patches); grounded variants with feature sets visual ($[\mathrm{subj}, \mathrm{ref}, \mathrm{subj} - \mathrm{ref}, \mathrm{subj} \cdot \mathrm{ref}]$), geometry (12-d normalized box features), and visual+geometry.

Ungrounded results (test acc %; balanced acc in parentheses; Wilson 95% CI; majority baselines: T1 64.2%, T2 64.7%, T3 49.6%):

T1 facing vs. away (n=103)	Test	Bal. acc	95% CI
ViT linear	65.0	60.4	[55, 74]
ViT MLP	55.3	52.0	[45, 65]
Merger linear	68.9	64.5	[59, 77]
Merger MLP	52.4	49.2	[42, 62]
ViT patch-vote	62.1	—	[53, 71]
Merger patch-vote	60.2	—	[51, 69]

T2 parallel vs. perpendicular (n=34)	Test	Bal. acc	95% CI
ViT linear	61.8	61.0	[44, 77]
ViT MLP	67.6	69.3	[50, 82]
Merger linear	64.7	61.4	[47, 79]
Merger MLP	64.7	63.3	[47, 79]
ViT patch-vote	73.5	—	[57, 85]
Merger patch-vote	67.6	—	[51, 81]

T3 4-way (n=137)	Test	Bal. acc	95% CI
ViT linear	52.6	40.5	[44, 61]
ViT MLP	52.6	42.9	[44, 61]
Merger linear	53.3	44.0	[45, 61]
Merger MLP	47.4	43.2	[39, 56]
ViT patch-vote	51.8	—	[44, 60]
Merger patch-vote	52.6	—	[44, 61]

Grounded results (CV acc / test acc; majority T1 63.7%, T2 57.0%; visual = $[\mathrm{subj}, \mathrm{ref}, \mathrm{subj} - \mathrm{ref}, \mathrm{subj} \cdot \mathrm{ref}]$; geometry = 12-d box features; T3 4-way is at chance for every readout, test acc 41.7–52.8% across all 12 cells vs. 49.9% majority, committed JSON):

Readout	T1 visual	T1 geom.	T2 visual	T2 geom.
ViT linear	59.8/61.6	61.1/60.6	51.2/53.6	44.2/50.0
ViT MLP	59.5/66.7	61.8/62.6	45.4/46.4	45.4/46.4
Merger linear	57.2/65.7	61.1/60.6	51.2/53.6	44.2/50.0
Merger MLP	56.3/52.5	61.1/71.7	49.0/53.6	50.0/42.9

The single above-majority cell (geometry-only merger MLP on T1, 71.7% test) has CV 61.1% below the 63.7% majority — not robust — and derives from the grounding model's own box statistics, not vision-feature content. In short, object-box conditioning with the model's own grounding predictions and mean-pooled region features does not recover robust orientation decodability under the tested readouts. The T2 patch-vote result (73.5%) is a suggestive point estimate with a wide CI overlapping the majority baseline (n=34).

Clear-subset control (canonical, frozen at paper-freeze-v1). The 48-example manual audit identified ambiguous or questionable orientation labels. Re-evaluation on progressively stricter subsets (values % from results/tables/orientation_clean_label_table.md, generated by scripts/clean_label_orientation.py): Because auditing changes the evaluated sample, strict-subset values are not a matched comparison against other relation families. The earlier probe-side clear/strict check (pre-freeze) additionally suggested that removing audit-flagged examples does not change the probe conclusion: the best ungrounded probe moved only 68.9% → 71.1% →

Table 2. Clean-label orientation robustness (canonical, frozen at paper-freeze-v1; values % from results/tables/orientation_clean_label_table.md).

Condition	Full (137)	−Questionable (132)	Clear (124)	Strict (107)
2B zero-shot	62.8	65.2	66.9	75.7
2B structured	53.3	52.3	51.6	53.3
2B General LoRA	62.0	63.6	64.5	72.9
2B Targeted LoRA	64.2	65.9	68.5	76.6
7B zero-shot	63.5	65.2	68.5	72.0
7B General LoRA	65.7	67.4	70.2	78.5
7B Targeted LoRA	64.2	65.9	69.4	72.0
7B Hard-Neg LoRA	66.4	68.2	70.2	76.6
7B Projector LoRA	64.2	65.9	69.4	73.8
7B Vision+Projector LoRA	64.2	65.2	68.5	73.8

71.6% (full/clear/strict, n = 103/97/88) and the generative T1 readouts 64.1/66.0/68.2 (zero-shot) and 68.9/72.2/78.4 (General LoRA); committed in results/probe/clear_subset_results.json.

4. Complete Statistical Test Battery

All pairwise tests are exact McNemar tests (binomial test on discordant pairs, two-sided). Discordant counts are reported as (condition-loses / condition-gains) relative to the control named in each row. No test was run and withheld; the battery below is complete for this paper.

Principal vs. exploratory comparisons. The battery above is fully enumerated; no test was run and withheld. The principal (confirmatory) comparisons are: zero-shot vs. LM-only consistency (ID 19) and the SITE transfer tests (§E). All other comparisons — vision-side conditions and their per-relation cells, hard-negative training vs. control, the remaining consistency comparisons, and the two-stage agreement rows (IDs 15–18) — are exploratory diagnostics reported with exact unadjusted p-values. Under a Holm adjustment over the 18 accuracy-valid VSR tests (α=0.05), only the zero-shot-vs-LM-only consistency test survives (Holm p=0.0018, using the conservative p<0.0001 bound); the vision-side overall effects (raw p=0.0043, p=0.0122; Holm 0.073, 0.195) and all other comparisons do not survive a blanket adjustment and are not load-bearing. For the 3 SITE transfer tests, Holm-adjusted p-values are 0.012 (primary), 1.0, 1.0.

Single-checkpoint note. Every trained condition is a single checkpoint. All McNemar tests above quantify paired disagreement on the fixed test examples between two conditions; they do not estimate training-run variability, and the single-seed results reported here do not estimate training-run variability. Adapter-rank claims (e.g., the 1–2pp overall differences among trained 7B conditions) should be read in that light.

Consistency verdict patterns. For each strict family and condition, the committed verdict table (results/consistency_verdict_table.json, scripts/consistency_verdict_table.py) reports the observed both-True / both-False / complementary counts, the original and flipped True rates, and the verdict rates expected under verdict independence given those base rates:

The zero-shot pattern is dominated by a False-defaulting response bias: expected consistency under verdict independence is 30.2%, below the observed 36.9%, so a simple base-rate null does not explain the facing inconsistency; conversely, training raises consistency while introducing both-True contradictions (0% → 24.3% LM-only / 17.5% hard-neg), a disclosed specialization trade-off. Full table for left/right (n=245), front/behind (n=314), and parallel/perp (soft, n=34) in the committed JSON.

Consistency definitions. Strict complements (exactly one truth value): left↔right (n=245), in front of↔behind (n=314), facing↔facing away (n=103). Soft complement (parallel↔perpendicular, n=34): both-False is legitimate; only both-True is a contradiction; both-True rates are reported separately. Self-consistency on a strict pair means opposite verdicts.

Wilson CIs. All per-relation and SITE subset accuracies are reported with Wilson 95% confidence intervals (main text and committed JSONs).

5. SITE Protocol, Keyword Rule, and Distributions

Preregistration. Protocol frozen 2026-08-09 before any evaluation; example IDs committed in results/site/site_protocol.json; config hash 28f4cc09887477af recorded in results/site/run_metadata.json. Evaluation-only: no SITE data used for training. Evaluation: Qwen2-VL-7B- Instruct frozen, greedy, official multiple-choice prompt and letter parser, max_new_tokens=16, images resized to ≤392 px long side (constant protocol

ID	Comparison	Subset	Discord.	p
1	HardNeg vs. Gen.	overall (2195)	52/60	0.508
2	HardNeg vs. Gen.	weak families	20/26	0.461
3	Projector vs. ctrl.	overall	—	0.0043
4	V+Proj vs. ctrl.	overall	—	0.0122
5	Projector vs. ctrl.	orientation	—	0.86
6	V+Proj vs. ctrl.	orientation	—	0.85
7–14	Proj./V+Proj vs. ctrl.	per-relation (4×2)	—	0.25–1.00
15	Two-stage A-lin. vs. ctrl.	agreement (127 boxed)	45/16	0.0003
16	Two-stage A-MLP vs. ctrl.	agreement (127 boxed)	47/14	<0.0001
17	Two-stage B-lin. vs. ctrl.	agreement (127 boxed)	44/20	0.0037
18	Two-stage B-MLP vs. ctrl.	agreement (127 boxed)	47/18	0.0004
19	Zero-shot vs. LM-only	consistency 662	117/43	<0.0001
20	HardNeg vs. LM-only	consistency 662	31/41	0.29
21	Projector vs. LM-only	consistency 662	63/48	0.18
22	V+Proj vs. LM-only	consistency 662	65/56	0.47

Table 3. Complete VSR test battery (22 rows). Per-relation cells (IDs 7–14) are enumerated in the committed vision_side_comparison.json: projector facing p=0.77, facing-away p=1.00, parallel p=0.25, perpendicular p=1.00; vision+proj facing p=1.00, facing-away p=0.58, parallel p=1.00, perpendicular p=1.00. Rows 15–18 are agreement-based: the committed two-stage pipeline scored agreement with the control's verdicts (not accuracy against ground truth), so these rows are descriptive and are excluded from the confirmatory set and from the Holm family. SITE transfer tests (3: all/primary/secondary, §E) form a separate small family.

Table 4. Facing-pair verdict patterns by condition (n=103); "exp." = expected under verdict independence given the observed base rates.

Condition (facing pairs, n=103)	both-T	both-F	comp.	orig-T	flip-T	exp. both-F	exp. comp.
7B zero-shot	0	65 (63.1%)	38 (36.9%)	0.20	0.17	66.5%	30.2%
LM-only LoRA	25 (24.3%)	10 (9.7%)	68 (66.0%)	0.62	0.52	18.0%	49.4%
HardNeg LoRA	18 (17.5%)	5 (4.9%)	80 (77.7%)	0.60	0.52	19.0%	50.2%
Projector LoRA	18 (17.5%)	15 (14.6%)	70 (68.0%)	0.53	0.50	23.5%	50.3%
Vision+Proj LoRA	14 (13.6%)	23 (22.3%)	66 (64.1%)	0.49	0.43	27.1%	49.4%

parameter; the installed transformers build ignores max_pixels), SDPA attention.

Subsets. Primary = official category spatial relationship reasoning (1,721 examples; 993 images). Secondary = heuristic orientation keywords (3,272; 2,024 images). Exploratory = official movement prediction & navigation (1,103; 248 images). Overlaps (image): primary ∩ secondary = 488, secondary ∩ exploratory = 186, primary ∩ exploratory = 0; union = 2,591 images, each evaluated once.

Exact orientation-keyword rule (applied, case-insensitive, word- boundary regex, to question text plus non-image option text; src/datasets/site.py):

orientation, oriented, orient,
facing, face, faces, direction,
directional, view, viewpoint,
view angle, angle, turned, rotate,
rotated, rotation, toward, towards,
away from, left, right, front,
behind, in front of, parallel,
perpendicular, clockwise,
counterclockwise

Secondary subset composition (n=2,024 images) by official category: multi-view & cross-image reasoning 866 (raw 37.8%, CAA 8.8%); spatial relationship reasoning 488 (71.3%, 53.5%); object localization & positioning 346 (46.5%, 28.3%); movement prediction & navigation 186 (49.5%, 24.3%); 3D information understanding 91 (27.5%, 0.4%); counting & existence 47 (53.2%, 37.1%). Top sources: exoego4d 316 (36.4%), egoexo4d 288 (34.7%), MMTBench 237 (48.9%), SAT 183 (46.4%), MMIU 133 (28.6%), CLEVR 106 (92.5%), SPEC 102 (50.0%), SpatialEval 100 (53.0%), ThreeDSRBench 95 (61.1%), MMERealWorld 89 (32.6%), SeedBench 63 (58.7%), GQA 41 (75.6%), VStarBench 37, Blink 35, plus 27 smaller sources.

Decision rule (preregistered). (1) If the primary subset shows the same broad weakness: run the VSR-trained 7B General LoRA on SITE (preregistered step 2). (2) Else if only the orientation subset shows the effect: narrow the claim to object/direction orientation. (3) Else: report benchmark-specificity. Outcome: branch 2; the claim was narrowed accordingly. The VSR-trained 7B General

LoRA adapter was subsequently evaluated on the identical SITE examples as a post-protocol cross-dataset transfer check (no SITE training; predictions in results/site/vsr_lora_predictions.csv, paired statistics from scripts/compare_site_zeroshot_lora.py, canonical table in results/tables/site_validation_table.md):

Entries are raw accuracy / CAA (%). The adapter is statistically neutral overall, does not improve the orientation subset, and significantly degrades the official spatial-relationship category — VSR gains transfer poorly and can induce negative transfer.

Composition-confound analysis (CPU-only, zero-shot predictions). Because the orientation slice is keyword-derived and heterogeneous, we test whether the heuristic flag predicts lower correctness after controlling for task composition (logistic regression over the 2,591 zero-shot predictions; scripts/site_confound_analysis.py → results/site/orientation_confound_analysis.json):

Within the official spatial relationship reasoning category, orientation-flagged questions remain descriptively 7.5pp harder than unflagged ones (71.3% vs. 78.8%; n=488/505). No source dataset has $n \geq 30$ in both arms (orientation-heavy corpora are almost entirely flagged, others almost never), so source indicators absorb most of the orientation variance in the full model. Reading: the adjusted association is in the expected direction but not significant; the low aggregate heuristic score is largely explained by task composition. We therefore do not treat SITE as independent confirmation of the VSR orientation bottleneck, and the main text reports the slice as exploratory.

High-precision post-hoc subset (exploratory only). Restricting the heuristic to VSR-construct terms only (facing, facing away, parallel, perpendicular; word-boundary match on question or option text) yields n=177 examples (77 in the official spatial-relationship category, 99 in multi-view/cross-image reasoning, 1 in object localization): raw accuracy 44.1% (95% CI $[37.0, 51.4]$), CAA -2.3%. This subset is explicitly post-hoc and does not replace the preregistered frozen definition; it is reported as exploratory only.

6. Failure Taxonomy

The 48 persistent-failure statements are the union of: wrong in both 7B zero-shot and 7B LoRA (20); wrong in $\geq$3 of 4 base conditions (38); wrong in both 2B and 7B zero-shot (28). A single-annotator exploratory audit (eight-class taxonomy) suggests many persistent errors remain visually interpretable while also identifying viewpoint and annotation ambiguities; it is qualitative evidence, not a causal localization of the failure. Annotations committed in results/failure_annotations.csv and results/orientation_persistent_annotations.csv.

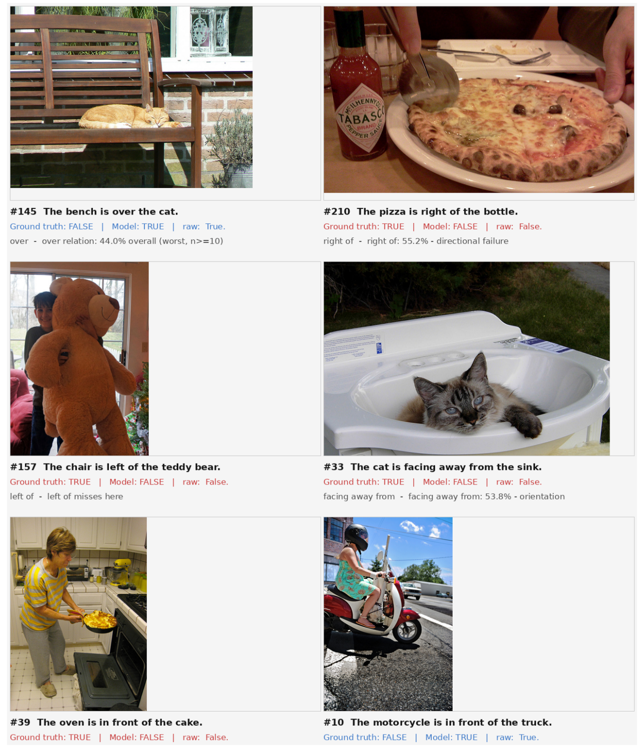

Figure 1. Representative persistent failures (annotated grid; 15 cases). Images: VSR test set. Per the single-annotator exploratory audit, the most frequent class is clear-image reasoning failure: the pose is visually interpretable to a human and the model still errs.

Transition counts. Scaling (2B→7B zero-shot): 28 both wrong, 23 2B-wrong/7B-right, 22 2B-right/7B-wrong, 64 both right. Adaptation (7B zero→7B LoRA): 20 both wrong, 30 fixed, 27 broken, 60 both right. Of the 48 annotated cases, hard-negative training fixes 5; 15 still fail under all three 7B conditions. Facing/facing-away account for 68.8% of persistent failures.

7. Reproducibility and Artifact Index

Code: anonymized archive accompanying this submission (no external links). Every headline number in this paper maps to a committed artifact; the audit script scripts/make_paper_figures.py recomputes the main tables directly from the raw prediction CSVs (output archived at paper/fig/audit_output.txt).

Environments. VSR/7B and SITE runs: Ubuntu 24.04, Python 3.12.13, torch 2.12.1+cu130, transformers 5.14.1, datasets 5.0.1, peft 0.20.0, 1× RTX A6000 (48 GB). 2B runs: earlier python3.10-era environment (SmolVLM2 wrapper). No lock files exist; the adapter

Table 5. SITE VSR-LoRA transfer check (post-protocol; predictions in results/site/vsr_lora_predictions.csv, canonical table in results/tables/site_validation_table.md). Subsets are defined by the frozen protocol IDs (see note below); no keyword reconstruction is used in analysis.

Subset	Zero-shot	VSR-LoRA	Δ	McNemar p
All images (n=2,591)	54.2 / 31.1	53.8 / 30.5	−0.4	0.66
Primary: spatial-rel. (n=993)	75.1 / 59.2	71.6 / 53.4	−3.5	0.004
Secondary: orient. kw. (n=2,024)	48.3 / 24.0	48.7 / 24.5	+0.4	0.70
Exploratory: movement (n=248)	48.8 / 23.1	50.4 / 25.6	+1.6	0.64

Table 6. Composition-confound logistic regressions (2,591 zero-shot SITE predictions; outcome = correct).

Model: correct ∼ orient + controls	OR (orient)	95% CI	p
Unadjusted	0.31	$[0.25, 0.38]$	2.4×10^{-28}
+ official category	0.74	$[0.57, 0.96]$	0.023
+ category, source, modality, $n_{options}$	0.84	$[0.60, 1.16]$	0.29

Failure mode	n	Share
Clear image, model reasoning failure	18	37.5%
Camera viewpoint ambiguity	8	16.7%
Parallel/perpendicular geometry	6	12.5%
Annotation questionable (label wrong)	5	10.4%
Intrinsic orientation ambiguous	4	8.3%
Front/back object ambiguous	4	8.3%
Small / occluded object	2	4.2%
Subject/reference inversion	1	2.1%

Table 7. Persistent-failure taxonomy, 48 annotated cases.

configs (committed) pin the LoRA hyperparameters. Seeds: 42 defaults in 2B paths; single default seed elsewhere — per-run seeds are not recorded in the metrics JSONs (a known limitation).

Canonical freeze. The experimental result set is frozen at tag paper-freeze-v1. Canonical tables are generated by scripts/build_canonical_tables.py; clean-label results by scripts/clean_label_orientation.py; publication figures by scripts/make_publication_figures.py; SITE paired transfer statistics by scripts/compare_site_zeroshot_lora.py; VSR/SITE headline recomputation by scripts/make_paper_figures.py (output archived at paper/fig/audit_output.txt).

Two-stage reproducibility caveat. The two-stage pipeline requires per-patch embeddings (results/probe/patch_embeddings.pkl, 8.2 GB, not committed; regenerable on GPU via scripts/extract_patch_embeddings.py) and hard-codes a Linux working directory. The committed evidence is the aggregate results/two_stage_results.json (agreement-based, see App. D); the per-example decisions are not committed.

SITE secondary-subset audit note (corrected before submission). An earlier derived analysis (compare_site_zeroshot_lora.py v1, which fed the initial zeroshot_image_metrics.json-adjacent figures and the first canonical site table) reconstructed the secondary subset by applying ORIENTATION_KEYWORDS to the question text only, yielding n=1,824 (47.3% raw / 22.6% CAA). The frozen protocol's preregistered definition — results/site/site_protocol.json → frozen_ids.secondary (3,272 examples; 2,024 with images), intersected with the 2,591 evaluated IDs — is the authoritative subset, and all numbers in this paper use it (n=2,024; zero-shot 48.3% / 24.0% CAA; VSR-LoRA 48.7% / 24.5%, p=0.70). The derived question-only figure was corrected before submission; both definitions support the same conclusion (orientation ≪ primary), but only the frozen-ID set matches the preregistration. No prediction CSV was modified and no model was rerun.

Data licenses/access. VSR: cambridgeltl/vsr_random (HF). SITE: franky-veteran/SITE-Bench, CC-BY-4.0 (media in 130 GB zips; requires HF read token). LoRA adapters are committed and loadable with peft; base-model weights via HF Hub.

Exact artifacts for the probe tables (App. 3): results/probe/probe_results.json, results/probe/patch_probe_results.json, results/probe/grounded_probe_results.json, results/probe/clear_subset_results.json, results/probe/grounded_boxes.json.

Headline number	Artifact	Commit
74.0 / 76.6 / 76.5 / 68.3 (2B)	smolvlm2_*_metrics_*.json	—
80.9 / 84.7 / 84.3 / 82.9 / 83.1 (7B)	*_metrics_*.json	—
62.8 / 63.5 / 65.7 / 66.4 (orient. family)	same + prediction CSVs	same
Per-relation cells (Table 1 (main))	orientation_analysis.json	—
48-failure taxonomy	orientation_persistent_*.json	—
Probe tables (App. 3)	probe/*.json	—
Two-stage (App. D, agreement-based rows 15–18)	two_stage_results.json	—
Consistency (Fig. 4 (main))	consistency_stats_all.json	—
Vision-side p-values	vision_side_comparison.json	—
SITE numbers	site/zeroshot_7b_predictions.csv site/zeroshot_image_metrics.json	—

Table 8. Headline-number → artifact → commit index (all paths relative to results/).